

Supplemental Material for “Transport-projected exact diagonalization of quantum spin ice: benchmarks and convergence criteria”

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(Dated: July 19, 2026)

MODEL, CONVENTION, AND CLUSTER GEOMETRY

We use local spin axes and

$$H = H_0 + V = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+). \quad (1)$$

Huang *et al.* use this same convention at the sign-free point $J_{\pm}/J_{zz} = 0.046$ [1]. No sign or factor-of-two conversion is therefore needed for the external benchmark.

Sites are represented in a periodic covering lattice. For each nearest neighbor pair (i, j) , the stored integer image $\mathbf{n}_{ij}$ gives the lifted bond displacement

$$\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij}^T L, \quad (2)$$

where the rows of L are the three supercell vectors. A depth-first cycle enumeration retains simple cycles that preserve the ice constraint and, for six-cycles, are planar in the covering lattice. The resulting census is

cluster	ice states	winding 4	contractible 6	wrapping 6
cubic-16	90	36	16	48
FCC-32	2970	24	32	96

There are no contractible four-cycles. The active geometry module performs the enumeration independently from the archived campaigns.

CANONICAL BLOCK TRANSFORMATION

Let P project onto the degenerate ice manifold of H_0 and $Q = 1 - P$. Introduce a bookkeeping parameter λ multiplying V . We seek

$$S^{(N)} = \sum_{n=1}^N \lambda^n S_n, \quad S_n^\dagger = -S_n, \quad PS_nP = QS_nQ = 0, \quad (3)$$

such that the off-diagonal block of $\tilde{H}^{(N)} = e^{-S^{(N)}} H e^{S^{(N)}}$ vanishes through order N . The last two conditions fix the canonical gauge [2].

For completeness, define $\mathcal{L}_0(X) = [H_0, X]$. Suppose orders below n have been fixed and let B_n be the coefficient of λ^n in the off-diagonal part of the Baker–Campbell–Hausdorff series with S_n omitted. Then

$$S_n = -\mathcal{L}_0^{-1}(B_n), \quad (4)$$

where the inverse acts only between P and Q . It exists while the Ising charge gap separating the chosen blocks remains nonzero. Equations (3) and (4) recursively define the series to arbitrary order; they are not a third-order ansatz.

The two finite-order blocks are

$$H_P^{(N)} = P \tilde{H}^{(N)} P, \quad (5)$$

$$H_Q^{(N)} = Q \tilde{H}^{(N)} Q, \quad (6)$$

and the omitted off-diagonal block is $O(\lambda^{N+1})$. An N -order calculation is therefore used only after comparing with $N + 1$ (or an exact band) over the temperature and frequency window reported.

At second and third order the row form used for the topology diagnostic is

$$H_P^{(2)} = PVRVP, \quad (7)$$

$$H_P^{(3)} = PVRVRVP, \quad R = -Q(H_0 - E_{\text{ice}})^{-1}Q. \quad (8)$$

The code stores each virtual sequence before rows with equal source, target, and image transport are merged. This is what permits a transport projection without inferring path content from the final matrix element.

TRANSPORTED DIPOLE AND CHARACTER PROJECTION

Consider a completed sequence that maps ice state $|a\rangle$ to $|b\rangle$. Every elementary $S_i^+ S_j^-$ transports one spin unit from j to i . Its lifted displacement is accumulated along the virtual sequence. In integer supercell coordinates the total is $\mathbf{q} \in \mathbb{Z}^3$. Equivalently, in Cartesian coordinates one may combine the home-cell dipole change and the accumulated bond images,

$$\delta_{ba} = \sum_{i \in \text{raised}} \mathbf{r}_i - \sum_{j \in \text{lowered}} \mathbf{r}_j + \mathbf{N}^T L. \quad (9)$$

Changing a site's home-cell representative shifts the two terms oppositely, so δ and the statement $\mathbf{q} = 0$ are gauge invariant.

Attach a phase $e^{i\boldsymbol{\theta} \cdot \Delta \mathbf{q}}$ to every elementary hop, where $\Delta \mathbf{q}$ is its lifted source increment. Individual increments may be half-integer, while every completed ice-to-ice path has integer $\mathbf{q}$. A completed row then has the Fourier expansion

$$[H_P^{(N)}(\boldsymbol{\theta})]_{ba} = \sum_{\mathbf{q}} h_{ba}^{(N)}(\mathbf{q}) e^{i\boldsymbol{\theta} \cdot \mathbf{q}}. \quad (10)$$

In the active exact calculation the primitive completed-path source is $\mathbf{q} = 2\delta$; the quarter-cell integer used by the row code is therefore $\delta_4 = 2\mathbf{q}$. This single normalization is used at every M . The discrete orthogonality identity

$$\frac{1}{M} \sum_{m=0}^{M-1} e^{2\pi i m q / M} = \begin{cases} 1, & q = 0 \pmod{M}, \\ 0, & \text{otherwise,} \end{cases} \quad (11)$$

proves

$$\mathcal{P}_M[H_P^{(N)}] = \sum_{\mathbf{q}=0 \pmod{M}} h^{(N)}(\mathbf{q}). \quad (12)$$

For fixed N only finitely many hops occur. Choosing M larger than every component of their possible net transport makes Eq. (12) the exact $\mathbf{q} = 0$ selector. At $N \rightarrow \infty$ arbitrarily large winding is possible, so a finite grid aliases multiple winding and M convergence is mandatory.

This explains two rejected alternatives. Averaging $\text{Tr } e^{-\beta H(\boldsymbol{\theta})}$ is nonlinear and does not define the partition function of the projected operator. Averaging microscopic Hamiltonians projects individual hops before they compose and therefore does not classify closed virtual paths. A fitted counterterm can cancel one leading matrix element, but it does not implement Eq. (12) at higher order or all temperatures.

EXACT ISOLATED-BAND LIMIT

Let P_λ project onto an isolated exact band with $\text{rank}(P_\lambda) = \text{rank}(P)$. If $PP_\lambda P$ is positive definite on P , the partial isometry

$$W_\lambda = P_\lambda P (PP_\lambda P)^{-1/2} \quad (13)$$

satisfies $W_\lambda^\dagger W_\lambda = P$ and $W_\lambda W_\lambda^\dagger = P_\lambda$. The pulled-back operator

$$H_{\text{band}} = W_\lambda^\dagger H W_\lambda \quad (14)$$

is the canonical all-order continuation of $H_P^{(N)}$ while the band remains isolated. Numerically, if Ψ contains the exact band eigenvectors and $X = P\Psi$, Eq. (13) becomes the Loewdin orthogonalization $X(X^\dagger X)^{-1/2}$. The minimum eigenvalue of $X^\dagger X$ is a required overlap diagnostic. A closing separation from the complement or a singular Gram matrix invalidates the selected-band protocol.

For every source angle θ , the same polar convention must be used before character averaging. Changing the arbitrary eigenvector gauge at each angle would otherwise produce a meaningless operator average. The active exact-band comparison uses this convention. At $J_\pm/J_{zz} = 0.046$, the centered changes are 6.78% from $M = 2$ to 3 and 0.498% from $M = 3$ to 4. The latter passes the 5% character-convergence gate. The zero-source minimum ice overlap is 0.762, and every generic $M = 3, 4$ point has a larger overlap.

FULL-TEMPERATURE THERMODYNAMICS

Let $\{E_i\}$ be the full microscopic spectrum, B the indices of the bare selected band, and $\{\tilde{E}_a\}$ the clean-band spectrum. Projection preserves the band trace; for a truncated series we enforce the same trace to fix the band's energy relative to the complement. Define, for $k = 0, 1, 2$,

$$M_k^{\text{clean}}(\beta) = \sum_{i \notin B} E_i^k e^{-\beta E_i} + \sum_a \tilde{E}_a^k e^{-\beta \tilde{E}_a}. \quad (15)$$

Then

$$E = M_1/M_0, \quad (16)$$

$$C = \beta^2 (M_2/M_0 - (M_1/M_0)^2), \quad (17)$$

$$S = \ln M_0 + \beta E. \quad (18)$$

Equation (15) is equivalent to diagonalizing one Hermitian operator whose selected block is replaced and whose complement is unchanged. For cubic-16 we use the complete 65,536-level spectrum. For FCC-32 the same identity requires stochastic trace estimation of the complement; that calculation has not yet been performed.

QMC BENCHMARK PROVENANCE AND METRICS

The reference heat capacity and entropy are vector extractions from Fig. 1(b) of Huang *et al.* [1]. The arXiv source contains the original vector PDF, allowing the red C/N and orange S/N paths to be read without raster tracing. The extraction script, axis calibration, CSV, and provenance are stored under `campaign/data/`. Estimated extraction uncertainties are 0.002 in $\log_{10}(T/J_{zz})$, 0.0015 in C/N , and 0.002 in S/N . These are not substitutes for unreported QMC statistical errors.

Curve error is the root-mean-square difference after interpolation in $\log T$ over the digitized support. Thresholds are defined in `campaign/config.json` before evaluation. The current values are

quantity	bare cubic-16	clean cubic-16	QMC
low T_p/J_{zz}	0.02554	0.001463	0.001100
low- T C/N RMSE	0.0859	0.02327	0
high T_p/J_{zz}	0.2549	0.2549	0.2105
S/N RMSE	0.0514	0.04509	0

The converged $M = 4$ result reduces the low-temperature heat-capacity RMSE by 72.9%, but its final value 0.02327 remains above the preregistered 0.02 tolerance. The entropy gate also fails because $0.04509 > 0.02$. The discrepancy is concentrated in the low-temperature region, where the finite clean band retains excess entropy relative to QMC.

The same paper specifies a future dynamics gate. It uses an $8 \times 8 \times 8$ primitive-cell lattice and reports

$$S^{zz}(\mathbf{q}, \omega) = \sum_{\alpha=1}^4 S_{\alpha\alpha}^{zz}(\mathbf{q}, \omega), \quad S^{+-}(\mathbf{q}, \omega) = \sum_{\alpha=1}^4 S_{\alpha\alpha}^{+-}(\mathbf{q}, \omega), \quad (19)$$

at $T/J_{zz} = 0.001, 0.04$, and 0.1 along $2\pi(0, 2k, 0)/8$ and $2\pi(k, k, k)/8$ [1]. ED comparison must use only momenta realized by its finite translation group, the same sublattice trace and spin normalization, a linear frequency axis, and a stated broadening comparable to the QMC-SAC resolution. This matched dynamics calculation has not yet been performed.

COMPLETE VALIDATION STATUS

gate	status	evidence or missing calculation
loop topology	pass	winding 4/6 removed; contractible 6 retained
all-temperature stability	pass	cubic-16 high- T peak position unchanged
QMC heat capacity	fail	exact $M = 4$ low- T RMSE $0.02327 > 0.02$
QMC entropy	fail	exact $M = 4$ RMSE $0.04509 > 0.02$
character resolution	pass	centered $M = 3 \rightarrow 4$ error 0.498%
band overlap	pass	source-resolved minimum is 0.762
order convergence	diagnostic only	production spectrum is exact-band
QMC-SAC dynamics	pending	matched S^{zz} and S^{+-} required
QMC energy	pending	not plotted in the benchmark paper
FCC-32 full Hamiltonian	pending	current result is order-three gauge block

The next production sequence is therefore fixed: include all conserved magnetization sectors in the unrestricted gap diagnostic; resolve the failed heat and entropy tests through cluster scaling; match the Huang *et al.* QMC-SAC spectra; and only then run the full-Hamiltonian FCC-32 and π -flux campaigns.

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- [1] C.-J. Huang, Y. Deng, Y. Wan, and Z. Y. Meng, Dynamics of topological excitations in a model quantum spin ice, *Phys. Rev. Lett.* **120**, 167202 (2018), [arXiv:1707.00099](#).
[2] S. Bravyi, D. P. DiVincenzo, and D. Loss, Schrieffer-wolff transformation for quantum many-body systems, *Ann. Phys.* **326**, 2793 (2011).